Three colours and three generations from perturbativity in 3-3-1 models

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Abstract

In $SU(3)_C \times SU(3)_L \times U(1)_X$ ("3-3-1") gauge models, anomaly cancellation ties the number of fermion generations to the number of colours. We keep the number of colours N_c general throughout, with every charge and anomaly coefficient computed as a function of N_c, and add a single requirement: no gauge coupling may become non-perturbative below the Planck scale. The $SU(3)_L$ anomaly together with asymptotic freedom of colour forces N_c to be odd and $N_g = N_c$, which makes Witten's global anomaly condition automatic. $SU(3)_L$ running then excludes $N_c \geq 7$ and $U(1)_X$ running excludes $N_c = 5$, so that $N_c = N_g = 3$ is the only survivor, with quark charges $\frac{2}{3}$ and $-\frac{1}{3}$. The selection holds at two loops in the gauge couplings with two-loop Yukawa terms, with one-loop threshold matching across the heavy spectrum, for 3-3-1 breaking scales from 2 to 30 TeV, and in both variants $\beta = \pm 1/\sqrt{3}$. It takes the measured electroweak couplings at the TeV scale as input and does not predict them; in the $\beta = +1/\sqrt{3}$ variant the $N_c = 3$ margin is a factor of about 2 in the pole scale. Requiring the scalar sector itself to remain perturbative and bounded below up to $M_{\rm Pl}$ then bounds the exotic-quark masses relative to the bilepton mass. All numbers are reproduced by open code with regression tests.

1 Introduction

In 3-3-1 models [1, 2] the $SU(3)_L$ anomaly cancels only between generations, which links the number of generations to the number of colours: N_g must be a multiple of N_c [3, 4, 5]. With the number of light neutrinos taken from experiment, Frampton noted that this singles out $N_c = 3$ [6]. Here no generation or neutrino number is taken from experiment. We ask instead which N_c is compatible with all gauge couplings remaining perturbative up to the Planck scale, given the measured electroweak couplings at the TeV scale. The answer is $N_c = N_g = 3$ only.

Section 2 sets up the model for general N_c. Section 3 gives the four-step selection; Sections 4–6 test its robustness, including one-loop threshold matching and the scalar sector. The conditions under which it holds are collected in Section 7, and Section 8 compares with prior work.

2 Setup for general N_c

The electric charge is $Q = T_3 + \beta T_8 + X$. The two variants without low-scale Landau poles are $\beta = \pm 1/\sqrt{3}$; the minimal $\beta = -\sqrt{3}$ model is excluded by LHC Z' limits (Section 9). For colour group $SU(N_c)$ we take N_g generations. Each has left-handed lepton triplets and quark triplets or anti-triplets, the third member of each quark multiplet being an exotic quark; n of the N_g quark generations sit in the anti-triplet.

Cancelling the perturbative anomalies, with the Yukawa couplings in place, gives

$$Q_u = \frac{N_c + 1}{2N_c}, \qquad Q_d = \frac{1 - N_c}{2N_c}, \tag{1}$$

which reduce to $\frac{2}{3}$ and $-\frac{1}{3}$ at $N_c = 3$. Every gauge anomaly (X^3, grav^2X, $SU(3)_L^2X$, $SU(N_c)^2X$, $SU(3)_L^3$) then vanishes identically in N_c, both in the Standard Model and in the full 3-3-1 content; this was checked in exact rational arithmetic for $N_c = 3, 5, 7$. Perturbative anomalies therefore fix the charges given N_c, but not N_c itself.

The $U(1)_X$ charges for $\beta = -1/\sqrt{3}$ are $X = Q_u - \frac{1}{3}$ (quark triplet), $Q_d + \frac{1}{3}$ (anti-triplet), $-\frac{1}{3}$ (leptons), and $-\frac{1}{3}, +\frac{2}{3}, -\frac{1}{3}$ for the scalars η, ρ, χ, fixed by the Yukawas and independent of N_c. For $\beta = +1/\sqrt{3}$: leptons $-\frac{2}{3}$, quark triplet $Q_u - \frac{2}{3}$, anti-triplet $Q_d + \frac{2}{3}$, scalars $\frac{1}{3}, \frac{1}{3}, -\frac{2}{3}$.

We require a non-abelian colour group, $N_c \geq 2$: for $N_c = 1$ there is no colour gauge field at all, and the asymptotic-freedom condition used below is vacuous.

3 The selection

Step 1: $N_g = N_c$. $SU(3)_L$ anomaly freedom requires equal numbers of triplets and anti-triplets,

$$N_g(1 + N_c) = 2nN_c. \tag{2}$$

For odd N_c, $\gcd(N_c, N_c + 1) = 1$, so $N_c \mid N_g$ and $N_g = kN_c$. With three quark flavours per generation, asymptotic freedom of $SU(N_c)$ requires $11N_c/3 - 2N_g > 0$, i.e. $k < 11/6$, so $k = 1$. For even N_c, $\gcd(N_c + 1, 2N_c) = 1$, so $2N_c \mid N_g$; but $2N_c > 11N_c/6$ violates asymptotic freedom. Hence N_c is odd and $N_g = N_c$, with no further input. At $N_c = 3$ this reproduces the standard assignment of two quark generations in the anti-triplet and one in the triplet.

Step 2: Witten's anomaly is implied. In the Standard Model, Witten's global $SU(2)$ anomaly [7] requires an even number of doublets, $N_c + 1$ per generation. In 3-3-1, $SU(2)_L \subset SU(3)_L$ and $\pi_4(SU(3)) = 0$, so the UV group imposes no global anomaly of its own. Step 1 already forces N_c odd, and the doublet count $N_g(1 + N_c) = 2nN_c$ is then even automatically. The spin–charge relation $(-1)^F = \exp(i\pi N_c(B - L))$ gives the same condition as a consequence rather than an input (all 29 solutions of (2) up to $N_c = 60$ checked).

Step 3: $SU(3)_L$ running excludes $N_c \geq 7$. With $N_g = N_c$ there are $N_c(1 + N_c)$ Weyl triplets. The one-loop coefficient $b_L = 11 - N_c(1 + N_c)/3 -$ (scalars), run from $1/\alpha_L(4\,\text{TeV}) = 31.5$, gives Table 1. Adding the minimal model's scalar sextet does not change the verdict.

N_c	b_L (3 scalar triplets)	Landau pole
3	$+6.5$	none (asymptotically free)
5	$+0.5$	none
7	-8.2	$1.3 \times 10^{14}\,\text{GeV}$, below M_{Pl}
9	-19.5	$10^8\,\text{GeV}$

Table 1: One-loop $SU(3)_L$ running with $N_g = N_c$.

Step 4: $U(1)_X$ running excludes $N_c = 5$. Two-loop gauge running from the 3-3-1 scale starts at $1/\alpha_X(4\,\text{TeV}) = 83.76$; we define "blow-up" as $g = 4\pi$. The results are in Table 2; the Planck scale is $1.2 \times 10^{19}\,\text{GeV}$. For $\beta = -1/\sqrt{3}$ a pole exactly at M_{Pl} needs $a_X = 14.76$, so the one-loop margins are $+6.09$ for $N_c = 3$ and -4.57 for $N_c = 5$. Two-loop gauge terms move the blow-up scale by a factor of about 3.

$N_c = N_g$	one-loop a_X ($\beta = -1/\sqrt{3}$; $+1/\sqrt{3}$)	blow-up, $\beta = -1/\sqrt{3}$	blow-up, $\beta = +1/\sqrt{3}$
3	$26/3 = 8.67$; 14.00	$\mathbf{3.0 \times 10^{29}}$ GeV	$\mathbf{4.0 \times 10^{19}}$ GeV
5	19.33; 27.33	1.3×10^{15} GeV	5.5×10^{11} GeV
7	34.44; 45.11	9.7×10^{9} GeV	3.1×10^{8} GeV

Table 2: Two-loop $U(1)_X$ blow-up scales.

Two-loop Yukawa terms. A Yukawa coupling $y\,\phi\psi_1\psi_2$ contributes $-g^3y^2\,n_{\text{terms}}\,[C_2(\psi_1) + C_2(\psi_2)]/d(G)/(16\pi^2)^2$ to each gauge β-function [8, 11]; this reproduces the Standard Model top-quark coefficients (17/10, 3/2, 2) exactly. We include the top Yukawa ($y_t = 0.83$ at 4 TeV) and one Yukawa y_J for every exotic quark, and for every heavy lepton when $\beta = +1/\sqrt{3}$, held constant up to the blow-up. Real Yukawas decrease with energy, so this overstates their effect and makes the $N_c = 5$ exclusion conservative. Yukawas raise every pole by at most a factor of about 5 (Table 3), and no constant $y_J \leq 6$ lifts $N_c = 5$ to the Planck scale.

y_J	$\beta = -1/\sqrt{3}$: $N_c = 3$ / $N_c = 5$	$\beta = +1/\sqrt{3}$: $N_c = 3$ / $N_c = 5$
no Yukawas	3.0×10^{29} / 1.3×10^{15} GeV	4.0×10^{19} / 5.5×10^{11} GeV
0 (top only)	3.4×10^{29} / 1.3×10^{15}	4.2×10^{19} / 5.6×10^{11}
1	4.7×10^{29} / 1.6×10^{15}	6.2×10^{19} / 6.8×10^{11}
2	1.2×10^{30} / 2.8×10^{15}	2.2×10^{20} / 1.3×10^{12}

Table 3: Blow-up scales with two-loop Yukawa contributions.

4 Robustness

The two-loop machinery reproduces the Standard Model one-loop coefficients $(41/10, -19/6, -7)$ and all nine two-loop coefficients exactly. Table 4 varies the 3-3-1 scale and the perturbativity criterion.

Check	$\beta = -1/\sqrt{3}$: $N_c = 3$ vs 5	$\beta = +1/\sqrt{3}$: $N_c = 3$ vs 5
3-3-1 scale 2–30 TeV	$(2.7\text{–}3.7) \times 10^{29}$ vs $(0.8\text{–}4.5) \times 10^{15}$	$(2.9\text{–}9.9) \times 10^{19}$ vs $(3.3\text{–}24) \times 10^{11}$
criterion $\alpha = 1$	1.8×10^{29} vs 1.0×10^{15}	2.9×10^{19} vs 4.7×10^{11}
criterion $g = 1$ (strict)	5.3×10^{25} vs 2.7×10^{13}	2.0×10^{17} vs 3.6×10^{10} ($N_c = 3$ fails)

Table 4: Blow-up scales in GeV under variations.

For $\beta = -1/\sqrt{3}$, $N_c = N_g = 3$ is selected under every criterion, with its blow-up 6–10 orders of magnitude above M_{Pl} while $N_c = 5$ fails 4–6 orders below. For $\beta = +1/\sqrt{3}$, 3 is selected under the standard criteria, but its pole lies only a factor 2–3 above M_{Pl} without Yukawas (3–5 with them); under the strict $g = 1$ criterion, $N_c = 3$ fails as well.

5 One-loop threshold matching

The breaking $SU(3)_L \times U(1)_X \to SU(2)_L \times U(1)_Y$ is matched at one loop in $\overline{\text{MS}}$ [12, 13]:

$$\frac{1}{\alpha_{\text{low}}(\mu)} = \frac{1}{\alpha_{\text{high}}(\mu)} - \frac{\lambda}{12\pi}, \quad \lambda = \sum_V t^2\Big(1 - 21\ln\frac{M_V}{\mu}\Big) + 8\sum_{\text{Dirac}} t^2 \ln\frac{M_F}{\mu} + \sum_{\text{real scalars}} t^2 \ln\frac{M_S}{\mu}, \quad (3)$$

with $1/\alpha_X + \beta^2/\alpha_L$ as the high-energy side for $U(1)_Y$. The heavy spectrum is: one complex $SU(2)$ doublet of bilepton vectors ($|Y| = \frac{1}{2}$, mass M_V); the Z' ($M_{Z'} = 1.225\,M_V$); N_c exotic

quarks (plus N_c heavy leptons for $\beta = +1/\sqrt{3}$) of mass $r_J M_V$, whose Yukawa $y_J = r_J g_L/\sqrt{2}$ enters at two loops; and nine heavy real scalars of mass $r_S M_V$ (a second doublet, a charged singlet and three neutral fields), with the option of a light second doublet.

As a check, at one loop the $N_c = 3$ pole must not depend on the matching scale once the spectrum is fixed: moving μ from $M_V/3$ to $3M_V$ changes ln(pole) by less than 10^{-9}, against a factor 2.6 without thresholds. $N_c = 5$ cannot pass this check, because the low-energy theory is the three-colour Standard Model; its pole moves by a factor 1.5 with μ, and we take its most favourable μ.

Scanning $r_J = 0.5$–3, $r_S = \frac{1}{3}$–3, $M_V = 2$–$10\,\text{TeV}$ and a heavy or light second doublet (two-loop running, Yukawas included) gives Table 5. The worst case for $N_c = 3$ (lightest exotics, light scalars, light second doublet, $M_V = 2\,\text{TeV}$) puts the Z' at 2.45 TeV, already below LHC limits. At two loops, moving μ from $M_V/2$ to $2M_V$ changes the pole by 1%.

	$N_c = 3$, worst case	$N_c = 5$, best case	verdict
$\beta = -1/\sqrt{3}$	$2.4 \times 10^{29}\,\text{GeV}$ $(2 \times 10^{10} M_{\text{Pl}})$	$7.2 \times 10^{15}\,\text{GeV}$	only $N_c = 3$
$\beta = +1/\sqrt{3}$	$2.0 \times 10^{19}\,\text{GeV}$ $(1.64\, M_{\text{Pl}})$	$3.8 \times 10^{12}\,\text{GeV}$	only $N_c = 3$
$\beta = +1/\sqrt{3}$, $M_V \geq 4\,\text{TeV}$	$2.23\, M_{\text{Pl}}$	—	only $N_c = 3$

Table 5: Selection with one-loop threshold matching over the scanned spectrum.

The selection holds for any cutoff Λ between the best $N_c = 5$ pole and the worst $N_c = 3$ pole. Over all criteria ($g = 4\pi$ or $\alpha = 1$, with or without Yukawas), any Λ from 7.2×10^{15} to $1.4 \times 10^{19}\,\text{GeV}$ works. This window contains M_{Pl}, the reduced Planck mass and the conventional unification scale.

6 The scalar sector to the Planck scale

The selection requires every coupling of 3-3-1 at $N_c = 3$ to stay perturbative to M_{Pl}, so the scalar sector must pass the same test. We implement the general one-loop quartic and Yukawa RGEs [9, 10, 11] on the 18 real scalars; the engine reproduces the Standard Model β_λ and β_{y_t} exactly. The nine quartics allowed by the model's discrete symmetry close under the RGEs (residual 10^{-15}), and the potential is gauge invariant to 10^{-16}. The light-Higgs quartic is matched at 4 TeV to $\lambda_{\text{SM}} = 0.076$ at tree level in the decoupling limit. Survival means every $|\lambda| < 4\pi$, every Yukawa $< \sqrt{4\pi}$, and a potential bounded below, checked numerically at eight scales up to M_{Pl}.

3-3-1 at $N_c = 3$ survives in both variants, in a corner of parameter space: 17 of 250 viable points ($\beta = -1/\sqrt{3}$) and 13 of 250 ($\beta = +1/\sqrt{3}$) under small priors, none with $O(1)$ quartics. In the survivors the extra scalars cure the Standard Model's metastability. Consistency to M_{Pl} then requires the exotic Yukawa at 4 TeV to be below 0.80 ($\beta = -1/\sqrt{3}$) or 0.62 ($\beta = +1/\sqrt{3}$), i.e. exotic-quark masses below $1.79\, M_V$ or $1.40\, M_V$; the top's doublet must carry $c^2 \gtrsim 0.68$ of the electroweak vev squared; and all quartics at 4 TeV must be $\lesssim 0.37$. These statements are one-loop, with tree-level matching, and the search is a random sampling rather than a proof that the surviving region is unique.

7 Conditions and caveats

1. *The framework.* Everything assumes 3-3-1 with $\beta = \pm 1/\sqrt{3}$.

2. *Perturbativity to the Planck scale* is a standard criterion, not a theorem. A pole below M_{Pl} signals new physics rather than an inconsistency.

3. *Measured couplings are input.* Every N_c is run from $\sin^2\theta_W = 0.2504$, $1/\alpha_{\rm em} = 125.7$ and α_s at 4 TeV; in 3-3-1, $1/\alpha_X = (1 - 4s^2/3)/\alpha_{\rm em}$. For $\beta = -1/\sqrt{3}$, $N_c = 3$ reaches $M_{\rm Pl}$ for $\sin^2\theta_W < 0.449$, while $N_c = 5$ does so only for $0.066 < \sin^2\theta_W < 0.074$, a factor 3.1–3.4 below the measured value. For $\beta = +1/\sqrt{3}$, $N_c = 5$ fails at every weak angle, and $N_c = 3$ survives for $\sin^2\theta_W < 0.266$ (tree-level matching; +0.010 to +0.035 in margin with thresholds). The precise statement is therefore: *given the measured electroweak couplings*, only $N_c = 3$ stays perturbative to $M_{\rm Pl}$.

4. *Minimality is not explained.* Steps 1–2 leave $\{3, 5, 7, \dots\}$; the running does the selecting.

5. *No numbers follow.* The structure adds about eleven scalar parameters beyond the Standard Model's two and blocks gauge unification ($SU(3)_C$ and $SU(3)_L$ never meet).

8 Relation to prior work

That N_g is a multiple of N_c in 3-3-1 is known [1, 2, 3, 4]. Frampton obtained $N_c = 3$ by inputting three light neutrinos [6]. Charge quantization for arbitrary N_c was discussed in [14], generation counting for arbitrary β in [15], and the N_c-dependence of the charges in [16]. The closest predecessor is Doff and Pisano [5], who keep n_c general and find $N_g = k\,n_c$ for any $k \geq 1$; they do not use asymptotic freedom to set $k = 1$, nor running to choose n_c. Landau poles in 3-3-1 at $N_c = 3$ are studied in [17, 18, 19].

The step taken here is to keep N_c general throughout (charges, anomalies, scalars, two-loop running, thresholds, scalar sector) without taking the number of generations or neutrinos from experiment. Searches of arXiv and INSPIRE, including every paper citing the founding 3-3-1 papers, Frampton [6] and Doff–Pisano [5], found no prior use of asymptotic freedom plus running to fix N_c. This is not proof of novelty, and comments from experts are welcome.

9 Experimental tests

The $\beta = \pm 1/\sqrt{3}$ models predict a Z', singly or doubly charged bilepton gauge bosons and exotic quarks, which the LHC and future colliders will find or exclude. Consistency to $M_{\rm Pl}$ adds the relations of Section 6: exotic quarks lighter than $1.79\,M_V$ or $1.40\,M_V$, $M_{Z'} = 1.225\,M_V$, and the top's doublet carrying $\gtrsim 70\%$ of the electroweak vev squared. A discovery violating these would break the selection without excluding 3-3-1. In the minimal $\beta = -\sqrt{3}$ model, $\sin^2\theta_W$ reaches $\frac{1}{4}$ at 3.68 TeV at one loop, forcing its Z' below about 3.7 TeV, in tension with LHC dilepton searches. The Z' mass limits and the low-energy R-ratio check of $N_c = 3$ are quoted from memory of the literature and have not been re-derived here.

Reproducibility

Each number above is produced by a script in `src/m331_*.py`, with reference output in `results/m331/` and regression tests in `tests/test_331_*.py`, run on every commit.

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